👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.NOS.6

Represent Rational Numbers on the Coordinate Plane

Lesson 7-5 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

LOST COVE MISSION

Captain Vega's Coordinates

You are the navigator on Captain Vega's expedition. Instead of 'X marks the spot,' she hid the treasure route as ordered pairs scrawled across the first quadrant of her chart, using only positive coordinates. Plot every point in the right order and the lost cove treasure is yours.

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can plot and identify points on the coordinate plane using ordered pairs.

Language Objective: I can explain my work using the words coordinate plane, ordered pair, origin, and quadrant.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Coordinate plane | **Ordered pair** | **Origin** | **Quadrant** | **Reflection** | **Integer** | **x-coordinate** | **x-axis** | **y-axis**

1. The grid made by a horizontal and a vertical number line is the _____.
2. Two numbers (x, y) that tell the location of a point are an _____.
3. The point (0, 0) where the x-axis and y-axis cross is the _____.
4. One of the four regions of the coordinate plane is a _____.
5. A flip of a point or shape over an axis is a _____.
6. A whole number or its opposite, with no fraction part, is an _____.
7. The first number in an ordered pair is the _____.
8. The horizontal number line on a coordinate plane is the _____.
9. The vertical number line on a coordinate plane is the _____.

Write About the Math

How do ordered pairs help the drone navigate? What shape does the flight path make?
¿Cómo ayudan los pares ordenados a que el dron navegue? ¿Qué forma hace la ruta de vuelo?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Coordinate plane** — *Plano cartesiano*
- ☐ **Ordered pair** — *Par ordenado*
- ☐ **Origin** — *Origen*
- ☐ **Quadrant** — *Cuadrante*
- ☐ **Reflection** — *Reflexión*

Model: *We start at the origin and move along the x-axis first because ___.* — Student states the x-coordinate is the sideways (horizontal) move and the y-coordinate is the up/down (vertical) move, both starting from the origin.

Start

Complete the frames. Use the word bank.
Completa las oraciones. Usa el banco de palabras.

- **The drone uses ordered pairs because the first number controls ___ movement and the second controls ___. The path makes a ___ shape.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.

- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Coordinate plane
2. **Notes 2:** Ordered pair
3. **Notes 3:** Origin
4. **Notes 4:** Quadrant
5. **Notes 5:** Reflection
6. **Notes 6:** Integer
7. **Notes 7:** x-coordinate
8. **Notes 8:** x-axis
9. **Notes 9:** y-axis
10. **Watch for this mistake:** *A common mistake in Graph on the Coordinate Plane is swapping the order of the coordinates — moving up/down first instead of right/left first. For example, when plotting (5, 2), students often move up 5 and right 2, landing on the wrong point (2, 5) instead of (5, 2). Remember: the x-coordinate (first number) always tells you how far to move right or left along the x-axis; the y-coordinate (second number) tells you how far up or down along the y-axis.*
11. **Try It 1:** A. Move 4 units to the right along the x-axis — *The first number in (x, y) is the x-coordinate, so 4 means move 4 units to the right along the x-axis before moving up.*
12. **Try It 2:** B. (3, 8) — *Across (3) is the x-coordinate and up (8) is the y-coordinate, so the Coconut Grove is at (3, 8).*